Bulacan Polytechnic College

Bachelor of Science in Information Systems

Bulihan, Malolos, Bulacan

“GradeTrack: A Mobile-Based Academic Progress Tracking Application”

A Project Documentation Presented to the Faculty of Bulacan Polytechnic College

In Partial Fulfillment of the Requirements for CT-MAD 213

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PROJECT OVERVIEW

1.1 Project Description

GradeTrack is a mobile-based academic progress tracking application designed to help second-year Bachelor of Science in Information Systems (BSIS) students of Bulacan Polytechnic College monitor and manage their academic progress. The application aims to provide an organized platform where students can access relevant academic information and keep track of their performance throughout the academic term.

The system has two primary user roles: Student, Faculty, and Admin. Students use the application to view and monitor their academic progress through the available features, while administrators, and faculty members manage the academic information and records supported by the system.

GradeTrack is intended to make academic monitoring more convenient, organized, and accessible. Through its features, the application seeks to help students become more aware of their academic standing and encourage them to take a more active role in managing their studies.

1.2 Problem Statement

Monitoring academic progress is an important part of a student's educational experience. However, students may experience difficulties keeping track of their academic performance and understanding their progress when relevant information is not presented in one organized platform.

The absence of a dedicated academic progress tracking application may make it more difficult for students to review their available academic information efficiently. Moreover, organizing and maintaining academic records may require additional effort from the individuals responsible for managing them.

Through this, GradeTrack proposes a mobile-based solution intended to support academic monitoring among second-year BSIS students at Bulacan Polytechnic College.

Specifically, the project seeks to address the following concerns:

1. How can students conveniently access and monitor their academic progress?
2. How can academic information be organized and presented in a clear and understandable manner?
3. How can administrators or faculty members manage the academic information supported by the application?

4. How can a mobile-based application help improve the accessibility and organization of academic progress information?

1.3 Target Users

GradeTrack is designed for the following users:

A. Student

The student is the primary user of the application. Students can access the academic information and monitoring features available to their accounts. The application aims to help them review their academic progress and become more aware of their performance.

B. Faculty

The faculty member is responsible for managing and updating students' academic records. This role includes viewing student lists, managing subject information, assigning grades for midterm and final assessments, and accessing class schedules. The faculty member ensures that students' grades are accurately recorded and made available for students to monitor their academic progress through GradeTrack.

C. Administrator

The administrator is responsible for managing and maintaining the overall system. This role includes managing student and faculty accounts, organizing academic information, and ensuring that the system operates properly. The administrator also oversees the accuracy and proper organization of the information stored in GradeTrack.

1.4 Proposed Solution

To address the identified concerns, the proponents propose GradeTrack, a mobile-based application that provides a centralized platform for academic progress monitoring.

The application is intended to organize relevant academic information and provide students with a convenient way to review their progress. It also provides an administrative interface for authorized personnel to manage the information required by the system.

Moreover, by integrating student access and academic information management into one application, GradeTrack aims to promote better organization, accessibility, and awareness of academic performance among second-year BSIS students at Bulacan Polytechnic College.

OBJECTIVES, SCOPE, AND FEATURES

2.1 General Objective

The general objective of the project is to develop GradeTrack, a mobile-based academic progress tracking application that helps second-year BSIS students of Bulacan Polytechnic College monitor their academic progress while providing authorized administrators and faculty members with tools to manage the academic information supported by the system.

2.2 Specific Objectives

Specifically, the project aims to:

1. Develop a mobile application that provides students with access to their available academic progress information.
2. Organize academic information in a clear, accessible, and user-friendly interface.
3. Provide appropriate user access and permissions for students and administrators or faculty members.
4. Enable authorized personnel to manage academic information according to their assigned responsibilities.
5. Evaluate the application's functionality, usability, and reliability through appropriate testing procedures.

2.3 Scope and Limitations

Scope

It focuses on academic progress monitoring for second-year BSIS students at Bulacan Polytechnic College. The application includes separate access for students, administrator, and faculty members, with features determined by each user's role.

The project covers the design and development of the mobile application's user interface, academic information management functions, and other implemented features related to monitoring student progress. It also includes testing to determine whether the application functions according to its intended requirements.

Limitations

The application is limited to the intended users, academic information, and functions included in the project. Its capabilities depend on the features actually implemented, the availability and accuracy of the information entered into the system, and the permissions assigned to each user role.

GradeTrack should not be described as an official college information system or as a replacement for existing institutional records unless formally authorized by the college. Features such as automated grade computation, notifications, attendance monitoring, and integration with existing school systems are outside the project's scope unless they are explicitly implemented.

2.4 Major Features

The following categories can be used to organize the application's features. Include only the functions that are present in the actual application.

Student Features

1. Account Access: Allows students to access the application through their assigned login credentials.

2. Academic Progress Dashboard: Presents a summary of the student's available academic information.

3. Academic Information Viewing: Allows students to view the academic records or progress details provided by the system.

4. Progress Monitoring: Helps students review their academic standing based on the information available in the application.

Faculty Features

1. Faculty Account Access: Provides faculty members with access to their assigned functions within the system.

2. Student List Viewing: Allows faculty members to view the list of students in their assigned section.

3. Student Grade Management: Enables faculty members to enter and update students' midterm and final grades.

4. Academic Records Management: Supports the accurate recording and maintenance of students' grades.

5. Grade Information Access: Allows faculty members to access the academic records of students under their assigned classes and subjects.

Admin Features

1. Administrative Account Access: Provides the administrator with access to the system's administrative functions.

2. Student Information Management: Allows the administrator to manage student information and accounts.

3. Faculty Account Management: Enables the administrator to manage faculty accounts and their access to the system.

4. Class Schedule Management: Allows the administrator to create, update, and delete class schedules for different sections.

5. Subject Management: Enables the administrator to add, edit, and delete subject information and assign instructors to subjects.

6. Academic Information Organization: Helps the administrator maintain organized student information, subject details, and class schedules.

General System Features

1. Role-Based Access: Separates the functions available to students and administrators or faculty members.

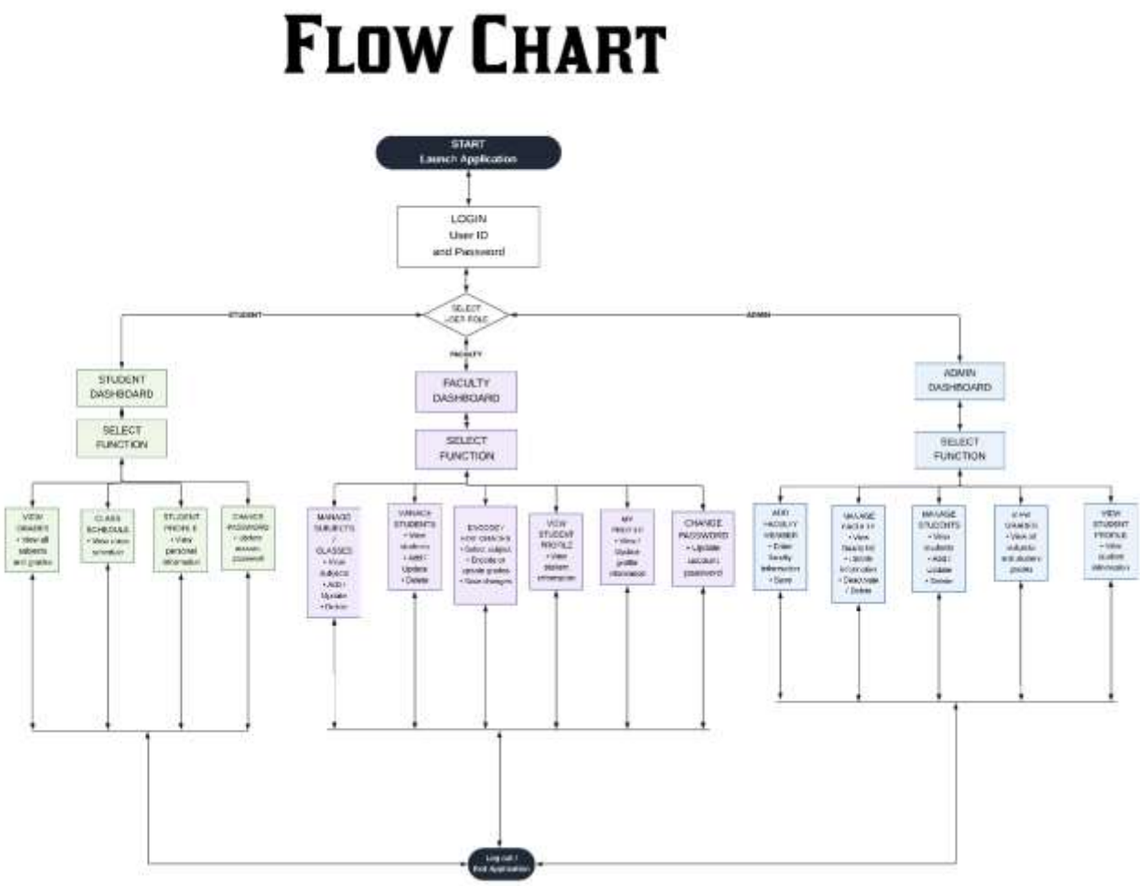
2. User-Friendly Interface: Provides a clear and consistent layout for navigating the application.

3. Data Management: Supports the storage, retrieval, and updating of information according to the application's implemented functions.

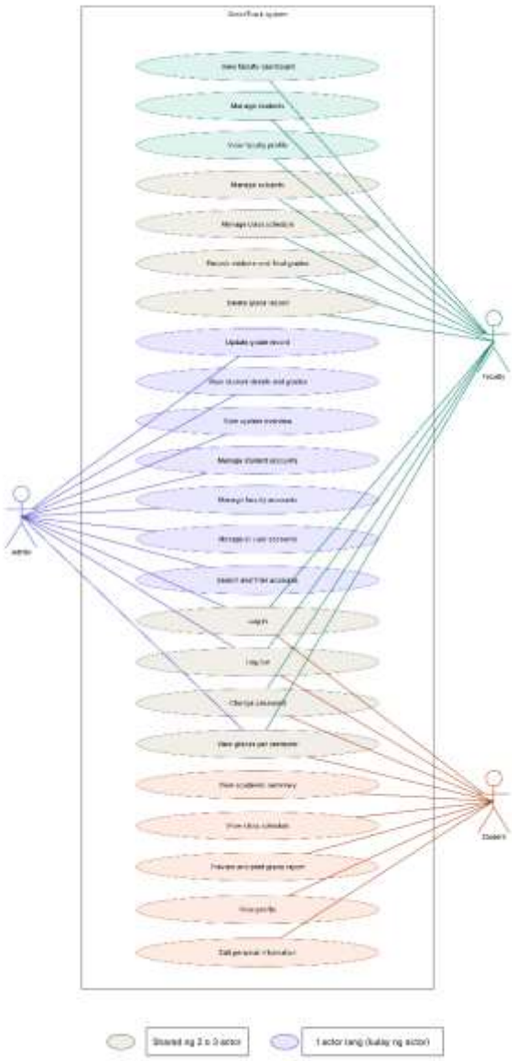
SYSTEM DESIGN

The system design of GradeTrack presents the overall structure and functionality of the application, including how users interact with the system and how academic information is managed. It describes the processes and components that support students and Admin and Faculty members in accessing and managing academic records. Students can view their profiles, class schedules, and grades, while Admin and Faculty members can manage student information, subjects, schedules, and grades according to their assigned permissions. The system design serves as a guide for understanding the application's workflow, user interfaces, data processing, and database organization to ensure that GradeTrack provides an organized and accessible way to monitor students' academic progress.

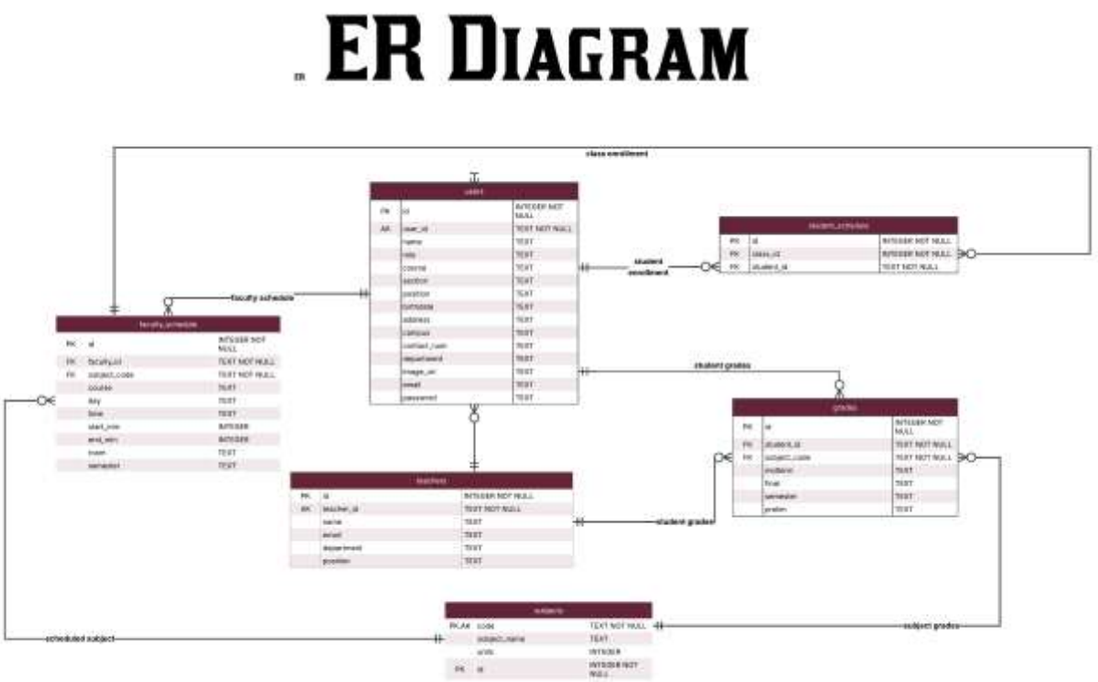
3.1 System/Application Architecture or Flow Diagram



3.2 Use Case Diagram



3.3. Entity Relationship Diagram (ERD)



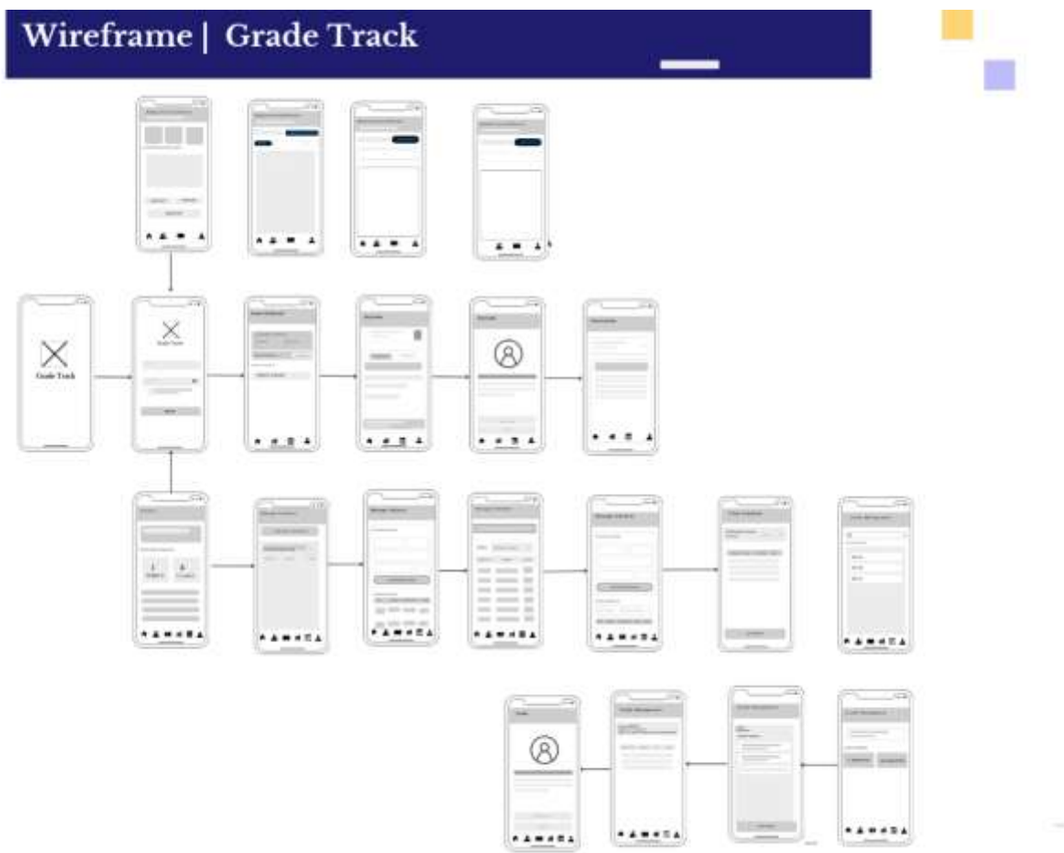
UI/UX AND IMPLEMENTATION

4.1 UI/UX Design

The user interface and user experience design of GradeTrack focus on clarity, consistency, and ease of navigation. The interface is intended to help students access academic progress information without unnecessary complexity.

The design should provide clearly labeled navigation elements, readable text, consistent layouts, and appropriate visual indicators for important academic information.

The student, Faculty, and Admin interfaces should reflect their respective responsibilities. Students should be able to locate their academic information easily, while authorized personnel should be able to access the management functions assigned to them.



4.2 Development Tools

1. Android Studio

Used as the primary development environment for writing code, designing the user interface, and building the GradeTrack application.

2. Laptop

Used as the main device for coding, designing, testing, and managing the application during development.

3. Android Emulator and Physical Android Devices

Used to test the application's interface, features, and functionality in simulated and real Android environments.

4. Claude AI

Used to assist with code generation, debugging, and programming suggestions during development.

5. Gemini AI

Used to explain programming concepts, troubleshoot technical issues, and suggest improvements to the application.

6. Notepad

Used to write and edit text files, code snippets, and development notes.

7. GitHub

Used to store and manage the source code, track project changes, and support collaboration among group members.

8. Lucidchart

Used to create system diagrams, such as flowcharts, use case diagrams, and other visual representations of the application.

9. Canva

Used to design presentation slides and other visual materials for the project documentation.

4.3 Technologies Used

1. Java

Used as a programming language to implement the application's functionality and logic.

2. Kotlin

Used as a programming language for developing Android application components and supporting mobile app functionality.

3. MySQL

Used as a database management system to store and manage student academic information and other relevant application data.

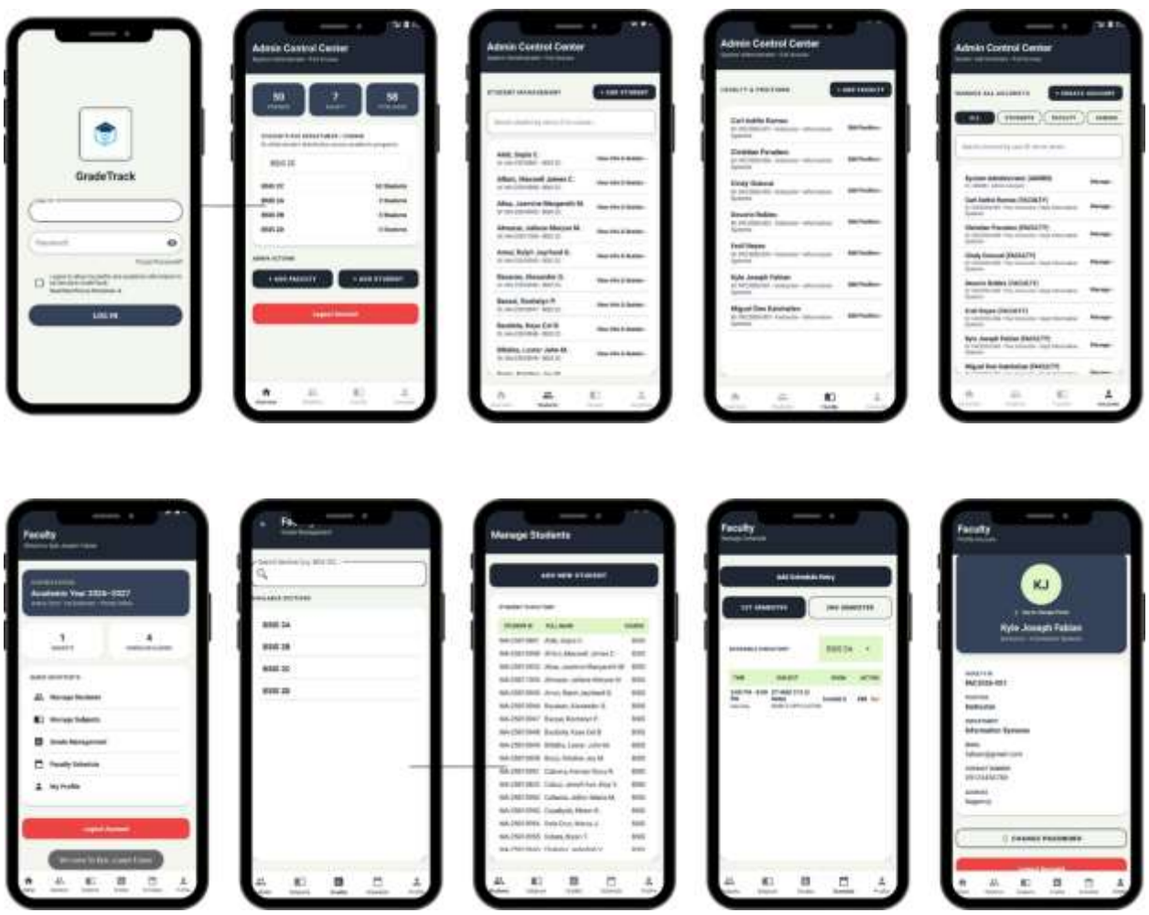
4. Android SDK

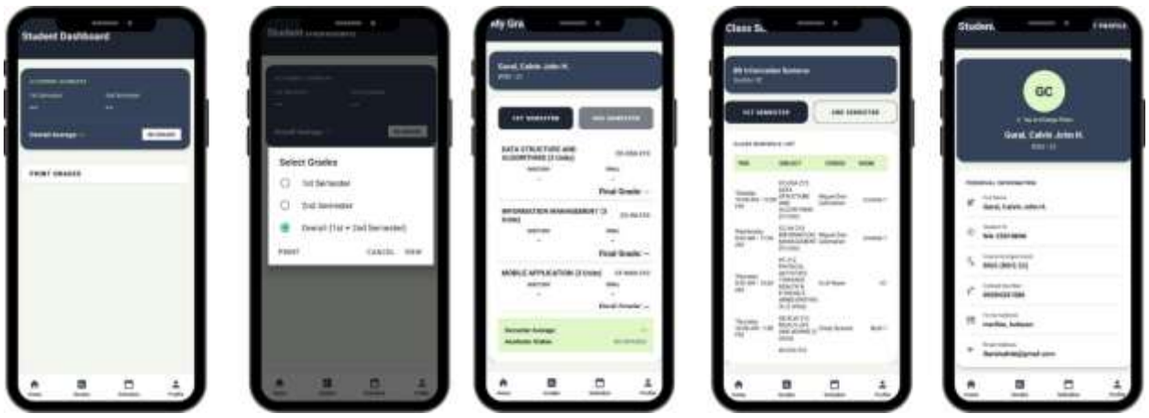
Used to provide the tools and libraries needed to develop and run Android applications.

5. XML (Extensible Markup Language)

XML was used to design the application's user interface, including buttons, text fields, financial cards, navigation components, and screen layouts.

4.4 Major Application Screens





4.5 Feature Implementation

1. User Login and Authentication

Allows students to log in to the application using their registered credentials and access the features available for their respective roles.

2. Student Dashboard

Provides an overview of the student's academic information, allowing users to access important details and navigate the application's main features.

3. Academic Progress Tracking

Allows students to view their academic performance and monitor their progress based on the academic information recorded in the system.

4. Grades Viewing

Enables students to view their recorded grades for their subjects, helping them understand their academic standing.

5. Student Information Management

Allows authorized Admin/Faculty users to manage student information and maintain accurate academic records.

6. Grade Management

Allows authorized Admin/Faculty users to add, update, and manage students' grades in the system.

7. Database Integration

Connects the application to its database to store, retrieve, and manage student information and academic records.

TESTING AND EVALUATION

5.1 Test Cases

The test cases were designed to evaluate the functionality of the GradeTrack application, including the Student Dashboard, Student Profile, Grade Track, Schedule Management, and Faculty & Admin modules. Each test case specifies the feature being tested, the steps to perform, and the expected result to determine whether the application functions according to its requirements.

Test Case ID	Feature	Test Steps	Expected Result
TC-01	Student Dashboard	Log in as a student and open the dashboard.	The dashboard displays the academic summary, overall average, academic status, and quick navigation options.
TC-02	Student Profile	Open the Profile page from the dashboard.	The student's personal information is displayed correctly.
TC-03	Grade Track	Open Grade Track and view the available subjects or academic information.	The available subjects and corresponding academic information are displayed.
TC-04	Academic Summary	View the academic summary on the Student Dashboard.	The semester averages, overall average, and academic status are displayed correctly.
TC-05	Schedule Management	Open Schedule from the dashboard.	The schedule directory and existing schedule entries are displayed.
TC-06	Add Schedule	Open Add Schedule Entry, fill in the required fields, and tap Add.	The new schedule entry is saved and appears in the schedule directory.
TC-07	Edit Schedule	Select Edit on a schedule entry, change its details, and save.	The updated schedule entry is displayed correctly.
TC-08	Delete Schedule	Select Del on a schedule entry.	The selected schedule entry is removed from the directory.
TC-09	Faculty & Admin Dashboard	Log in as Admin/Faculty and open the dashboard.	The academic year, active semester, and management modules are displayed.
TC-10	Student Management	Open Manage Students and perform a supported student management operation.	The student management feature works as expected.
TC-11	Subject Management	Open Manage Subjects and perform a supported subject management operation.	The subject management feature works as expected.
TC-12	Grade Management	Open Manage Grades and perform a supported grade management operation.	The grade management feature works as expected.
TC-13	Semester Selection	Open Schedule Management and switch between semesters.	The schedule directory displays entries for the selected semester.
TC-14	Section Search	Open Grade Track, enter a section name in the search field, and check the results.	The application displays sections matching the search input.
TC-15	Logout	Tap Logout Account on the dashboard.	The user is logged out and returned to the login screen.

5.2 Test Results

The testing results present the actual outcomes of the tests performed on the GradeTrack application. All 15 test cases passed, indicating that the implemented features worked as expected during

testing. The results demonstrate that the application can display student information and academic records, manage schedules, support administrative functions, and provide essential navigation features.

Test Case ID	Feature Tested	Actual Result	Status
TC-01	Student Dashboard	Dashboard details and navigation options displayed correctly.	Passed
TC-02	Student Profile	Student personal information displayed correctly.	Passed
TC-03	Grade Track	Available subjects and academic information displayed correctly.	Passed
TC-04	Academic Summary	Academic averages and status displayed correctly.	Passed
TC-05	Schedule Management	Schedule directory and entries displayed correctly.	Passed
TC-06	Add Schedule	New schedule entry was added successfully.	Passed
TC-07	Edit Schedule	Schedule entry was updated successfully.	Passed
TC-08	Delete Schedule	Selected schedule entry was deleted successfully.	Passed
TC-09	Faculty & Admin Dashboard	Academic year and management modules displayed correctly.	Passed
TC-10	Student Management	Student management feature worked as expected.	Passed
TC-11	Subject Management	Subject management feature worked as expected.	Passed
TC-12	Grade Management	Grade management feature worked as expected.	Passed
TC-13	Semester Selection	Schedule entries updated for the selected semester.	Passed
TC-14	Section Search	Section search returned appropriate results.	Passed
TC-15	Logout	User logged out and returned to the login screen.	Passed

Reported Testing Summary

- Total tests Cases: 5
- Passed: 5
- Failed: 0
- Reported Pass Rate: 100%

The GradeTrack application underwent functional testing to evaluate the performance of its major features. The testing covered the Student Dashboard, Student Profile, Grade Track, Academic Summary, Schedule Management, and Faculty & Admin modules. The results showed that all 15 test cases passed, with each feature functioning according to its expected behavior. The application successfully displayed student information and academic records, managed schedules, provided access to student, subject, and grade management functions, and supported section searching, semester selection, and account logout. Based on the test results, the implemented features performed as intended during testing.

5.4 User Evaluation

The user evaluation of GradeTrack: A Mobile-Based Academic Progress Tracking Application aims to assess the application's usability, interface design, navigation, functionality, and usefulness to its intended

users, particularly students, faculty, and Admin. The evaluation focuses on determining whether the application is easy to understand and operate, provides a clear and consistent interface, allows users to access its main features efficiently, and performs its functions according to user expectations. It also examines how effectively GradeTrack helps students monitor their academic performance, view their grades, and track their academic progress. The feedback gathered from users can help identify areas for improvement and ensure that the application meets the needs of its intended users.

Evaluation Criteria

1. Ease of Use: Evaluates how easily users can understand and operate the application, including accessing features and viewing academic information.

2. Interface Design: Assesses the clarity, consistency, organization, and visual appearance of the application's interface.

3. Navigation: Determines how easily users can access the application's main features, such as the Student Dashboard, Grade Track, Student Profile, and Schedule.

4. Functionality: Evaluates whether the application's features work according to user expectations, including viewing grades, managing schedules, and handling student and academic records.

5. Usefulness: Measures how effectively GradeTrack helps students monitor their academic performance, review their grades, and keep track of their academic progress.

Testing Conclusion

The testing covered the Student Dashboard, Student Profile, Grade Track, Academic Summary, Schedule Management, and Faculty & Admin modules. Based on the testing results, all 15 test cases passed, indicating that the implemented features worked according to their expected behavior during testing. The application successfully displayed student information and academic records, supported schedule management, and provided access to administrative functions. Overall, the testing demonstrated that GradeTrack meets its intended functional requirements and can help students monitor their academic performance and progress. Further improvements may be made based on user feedback to enhance the application's usability, interface design, and overall user experience.

CONCLUSION AND RECOMMENDATIONS

6.1 Project Accomplishments

The development of GradeTrack: A Mobile-Based Academic Progress Tracking Application successfully established a mobile platform designed to help students monitor their academic performance and progress. The application includes essential features such as the Student Dashboard, Student Profile, Grade Track, Academic Summary, Schedule Management, and Faculty & Admin modules. It allows students to view their personal information, access academic records, and monitor their grades, while Admin/Faculty can access management features for student information, subjects, and grades. The application also underwent functional testing, in which all 15 test cases passed based on the testing conducted. These accomplishments demonstrate that the project achieved its primary objective of providing a more organized and accessible way to monitor academic progress.

6.2 Problems Encountered

During the development of GradeTrack, the team encountered several challenges in implementing the application's features and ensuring that the different modules worked properly. Some of the problems encountered included:

- **Feature Implementation:** Developing the application's functions and ensuring that each module worked as intended.
- **Coding Errors:** Identifying and fixing errors during the development process.
- **Data Management:** Ensuring the accuracy and proper handling of student information, subjects, and academic records.
- **User Interface Design:** Organizing the layout to make the application clear and easy to use for both students and Admin/Faculty.
- **Navigation and Integration:** Ensuring that users could navigate between screens and that the different modules worked together properly.

6.3 Recommendations and Future Improvements

GradeTrack may be improved in the future by adding features that provide students with a more comprehensive view of their academic performance. The following improvements are recommended:

- **Push Notifications:** Add reminders for class schedules, academic deadlines, and important school activities.
- **Grade Visualization:** Include charts and graphs to help students understand their academic performance and grade trends.
- **Academic Reports:** Provide downloadable or printable reports of students' grades and academic progress.
- **Improved Data Security:** Strengthen user authentication and protect students' personal information and academic records.
- **Database Backup**: Implement regular backups to prevent the loss of important academic data.
- **Enhanced User Interface:** Improve the application's design, readability, and accessibility for a better user experience.
- **Performance Optimization:** Improve loading times and overall application responsiveness.
- **User Evaluation**: Gather feedback from more students and faculty members to identify areas that need improvement.

APPENDICES

The appendices contain supplementary materials that support the development, implementation, and evaluation of GradeTrack.

Appendix A. Testing and Evaluation Forms

FINAL SCORE SUMMARY (Include this in your documentation)

Criteria	Maximum Score	
A. Functionality & Features	25	____
B. UI/UX Design & Usability	20	____
C. Technical Implementation	20	____
D. Testing & Reliability	10	____
E. GitHub & Source Code Management	10	____
F. Documentation	5	____
G. Presentation & Demonstration	10	____
TOTAL	**100**	**____ / 100**

Final Rating

Raw Score: _____ / 100

Equivalent Grade: _____

Evaluator's Overall Comments:

XIX. PROJECT EVALUATION DECISION

☐ **APPROVED** – Project meets the required standards.

☐ **APPROVED WITH MINOR REVISIONS** – Project is acceptable but requires minor corrections.

☐ **FOR REVISION** – Significant improvements are required before final acceptance.

☐ **NOT ACCEPTED** – Project does not meet the minimum project requirements.

Required Revisions:

1.

2.

3.

4.

XX. EVALUATOR SIGNATURE

Evaluated by:

Name: _____

Signature: _____

Date: _____

Appendix B. GitHub Repository

Account: guralcalvin82

Link: https://github.com/guralcalvin82-dev/GradeTrack/tree/master